Session 2 Worksheet: Tame Architectures and Learnability

Georgia Workshop on Logic and Artificial Intelligence

1. **Unions of a bounded number of intervals.**

   Fix $k \geq 1$, and let $\mathcal{U}_k$ be the class of subsets of $\mathbb{R}$ that are unions of at most k open intervals.

   (a) Write a formula
   $$\varphi_k(x; \bar{a}, \bar{b})$$
   in the language of ordered fields whose parameter instances define exactly the members of $\mathcal{U}_k$.

   (b) Let
   $$x_1 < \cdots < x_{2k}.$$
   Prove that $\{x_1, \ldots, x_{2k}\}$ is shattered by $\mathcal{U}_k$.

   (c) Let
   $$x_1 < \cdots < x_{2k+1}.$$
   Prove that this set is not shattered by considering the alternating labelling
   $$1, 0, 1, 0, \ldots, 0, 1.$$

   (d) Determine $\mathrm{VCdim}(\mathcal{U}_k)$.

2. **Self-attention and o-minimal definability.**

   Fix a sequence length n, an input dimension d, and a head dimension h. An input is a sequence
   $$X = (x_1, \ldots, x_n) \in (\mathbb{R}^d)^n.$$
   Let
   $$W_Q, W_K, W_V \in \mathbb{R}^{h \times d}$$
   be parameter matrices. For $i = 1, \ldots, n$, define
   $$q_i = W_Q x_i, \qquad k_i = W_K x_i, \qquad v_i = W_V x_i,$$
   so that $q_i, k_i, v_i \in \mathbb{R}^h$.

   The attention weights and output vectors are
   $$\alpha_{ij} = \frac{\exp\Big(q_i \cdot k_j / \sqrt{h}\Big)}{\displaystyle\sum_{\ell=1}^{n} \exp\Big(q_i \cdot k_\ell / \sqrt{h}\Big)}, \qquad z_i = \sum_{j=1}^{n} \alpha_{ij} v_j.$$

(a) Prove that the joint map

$$(X, W_Q, W_K, W_V) \longmapsto (z_1, \ldots, z_n)$$

is definable in

$$\mathbb{R}_{\exp} = (\mathbb{R}, <, +, \cdot, \exp).$$

(b) A fixed multi-head attention layer consists of finitely many heads of the form above, followed by concatenation and an affine map. A fixed transformer architecture may also use finitely many such layers, residual additions, and coordinatewise ReLU feedforward maps.

Explain why the joint input-parameter map of any such fixed architecture is definable in $\mathbb{R}_{\exp}$.

(c) Add parameters $u \in \mathbb{R}^h$ and $b \in \mathbb{R}$, and define the binary classifier

$$X \longmapsto \mathbf{1}\{u \cdot z_1 + b \geq 0\}.$$

Write down the resulting concept class as the parameters vary. Explain why it has finite VC dimension and is distribution-free PAC learnable.

Which quantities in the description of the architecture must remain fixed for this argument to produce one uniformly definable family?

3. **Yarotsky's superexpressive architectures.**

Call a fixed one-input architecture

$$F_\theta : [0,1] \longrightarrow \mathbb{R}$$

superexpressive if, for every $f \in C([0,1])$ and every $\eta > 0$, some choice of weights θ satisfies

$$\sup_{x \in [0,1]} |F_\theta(x) - f(x)| < \eta.$$

Yarotsky gives superexpressive architectures using the activation family $\{\sin, \arcsin\}$ and also constructs a single C^1, bounded, strictly increasing superexpressive activation σ_3.

(a) Let

$$0 < x_1 < \cdots < x_m < 1$$

and prescribe arbitrary signs

$$\epsilon_i \in \{-1, +1\}.$$

Construct a continuous function f satisfying

$$f(x_i) = \epsilon_i \qquad (i = 1, \ldots, m),$$

and use superexpressiveness with approximation error less than $1/2$ to show that the threshold class

$$\mathcal{H} = \big\{\{x \in [0,1] : F_\theta(x) \geq 0\} : \theta\big\}$$

realizes the prescribed labelling.

(b) Deduce $\mathrm{VCdim}(\mathcal{H})$.

(c) Suppose that all activation functions used by this fixed architecture were definable in one common o-minimal expansion of the real field. Derive a contradiction.

State separately what follows for a finite superexpressive activation family and what follows for a single superexpressive activation such as σ_3.

(d) Give a direct reason that global sine is not definable in any o-minimal expansion of the real field.

(e) Explain why these conclusions do not contradict ordinary universal approximation theorems for ReLU networks.